PRIYA BALAKRISHNAN

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EDUCATION

The University of Massachusetts Amherst **Amherst, MA**
Master of Science in Computer Science - ML / AI concentration (**GPA 3.8 / 4.0**) *Anticipated - Sept 2024 - May 2026*

- Relevant Courses**:** Statistics, Neural Networks, Systems for Data Science, Natural Language Processing, Computer Vision, Software Engineering, Optimization in Machine Learning.

The University of Queensland **Brisbane, Australia**
Bachelor of Engineering (Hons) - Electrical and Computer Engineering (**GPA 4.7 / 7.0**) *Jul 2018 - Jul 2023*

- Relevant Courses**:** Data structures and Algorithms, Object Oriented Programming, Machine learning, Operating Systems, Computer Architecture and Organization, Web Design and Development.

PUBLICATIONS

Visualising Physiological Cues in Individualised Virtual Reality for Mental Health *Oct 2023*

- Presented at the **22nd IEEE International Symposium on Mixed and Augmented Reality (ISMAR)** as the **Second Author** under the guidance of Dr. Nilufar Baghaei from the School of Electrical Engineering and Computer Science.
- **Independently designed and developed** four distinct VR visualizations for **physiological data representation**, enhancing VR's therapeutic potential in mental health interventions.
- **Created expert and user testing plans and conducted usability testing with 30+ participants**, refining visualizations based on feedback to enhance effectiveness and identifying the most preferred design for improved engagement.
- **Conducted a literature review, defined design principles, and authored the research paper**, contributing to the scalable application of **individualized VR solutions** for mental health treatment

PROFESSIONAL EXPERIENCE

Deloitte Digital **Brisbane, Australia**
Graduate Full-stack Software Developer *Jul 2023 - May 2024*

- **Developed the user authentication system by integrating Auth0 identity management** for Travel Money Oz, Australia's largest foreign exchange provider, securing access for **100,000+ users** transacting currency online.
- **Optimized customer authentication** by implementing Multi-Factor Authentication (MFA) and integrating mobile phone, passport, and ID verification, **reducing authentication-related support issues by ~70%** and enhancing security for same-day currency delivery.
- **Automated migration of 10,000+ user records** from a SQL Server database to Auth0 using C# encryption and decryption, completing the process in minutes and **reducing manual effort by 90%.**
- **Developed a dashboard application** for employees using ASP.NET RESTful APIs and a React.js (TypeScript) front-end, enabling staff to monitor exchange rates and manage customer data with real-time SQL database updates.
- **Interacted directly with stakeholders** to understand business requirements and user challenges, translating insights into solution design and implementation.

Deloitte Digital **Brisbane, Australia**
Summer Intern - Backend Software Developer *Aug 2021 – Dec 2021*

- **Engineered a multi-currency payment feature** for **Travel Money Oz** using **ASP.NET and C#**, enabling users to add multiple currencies to travel cards seamlessly, improving **transaction efficiency by 25%** and boosting customer satisfaction.
- **Developed backend APIs for real-time currency conversion and secure payment processing**, ensuring accuracy and scalability.

SKILLS

Python Libraries and Frameworks: sckit-learn, pandas, numpy, keras, PyTorch, pySerial, paho-mqtt.
Programming Languages: Python, C#, C++, Java, SQL, JSON, YAML files, JavaScript, R, Matlab, Objective-C, JavaScript.
Frameworks and Libraries: .NET Framework, ASP.NET Web API, React.js, TypeScript, Node.js
Tools and Platforms: Linux, Git, Postman, Jira, MySQL, Tableau, Bash, Docker, Jenkins, Kubernetes, Oracle.

SELECTED PROJECTS

Fairness-Aware Facial Recognition Models Using the Seldonian Framework *Sep 2024 - Dec 2024*

- Implemented a **fairness-aware facial recognition model** using the **Seldonian frameworks**, **reducing racial bias by 20%** in age prediction tasks, while maintaining equitable accuracy across demographic groups.
- Designed a **constrained CNN model** using Pytorch and scikit-learn and analyzed fairness metrics using the **FairFace dataset**, achieving **95% compliance with fairness constraints** while maintaining high accuracy and low constraint violation probability, demonstrating superior fairness compliance over baseline models like ResNet50 and VGG16.

Indoor Air Quality Indicator with Mechanical Plant Avatar *Feb 2023 - Jun 2023*

- Directed firmware development for an Indoor Air Quality Indicator, integrating environmental sensors to monitor air quality and enabling **BLE-based real-time IoT data streaming** for predictive air quality monitoring.
- Built a traditional ML-based air quality classifier using sensor data, achieving **80% accuracy** in condition classification.